

HAINES, AK, USA

WMO: 703626

Lat: 59.243N

Lon: 135.509W

Elev: 15

StdP: 14.69

Time Zone: -9.00 (NAK)

Period: 98-19

WBAN: 25323

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.8	5.4	-13.9	2.6	6.3	-8.7	3.4	8.9	24.0	27.3	21.7	25.1	12.6	290	0.693

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	73.8	58.7	69.8	57.7	66.6	56.5	60.8	70.7	59.4	67.2	58.1	64.3	8.0	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.9	69.1	62.6	55.8	66.3	61.4	54.8	63.9	60.3	26.9	71.0	25.9	67.4	25.0	64.2	67.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.5	18.8	17.5	-3.2	81.5	4.2	4.5	-6.2	84.8	-8.7	87.4	-11.0	89.9	-14.0	93.2	
			DB												
			WB	-4.2	63.0	4.1	1.9	-7.1	64.4	-9.5	65.6	-11.8	66.6	-14.7	68.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.6	25.9	27.5	32.0	41.1	50.2	55.9	58.1	57.1	50.9	42.4	31.0	26.6
	DBStd	13.94	10.75	9.12	8.22	5.08	5.18	4.36	3.62	3.73	3.78	4.93	8.91	9.87
	HDD50	3834	748	630	559	271	61	2	0	0	31	237	569	726
	HDD65	8538	1213	1050	1024	716	459	276	219	249	422	700	1019	1191
	CDD50	782	0	0	0	5	67	180	250	219	59	2	0	0
	CDD65	12	0	0	0	0	0	5	4	3	0	0	0	0
	CDH74	118	0	0	0	0	4	48	33	33	0	0	0	0
	CDH80	16	0	0	0	0	0	6	5	5	0	0	0	0
Wind	WSAvg	9.3	11.2	10.9	10.5	9.2	8.3	7.4	7.1	7.2	8.2	9.5	10.9	11.5
Precipitation	PrecAvg	47.9	4.9	4.7	3.5	2.6	1.9	1.5	1.6	2.6	5.6	9.4	5.8	5.6
	PrecMax	59.7	10.9	10.4	6.6	6.1	7.2	5.2	3.3	7.0	12.2	20.4	13.7	12.8
	PrecMin	36.0	1.0	0.0	0.0	0.4	0.2	0.2	0.1	0.0	0.4	4.1	1.5	0.7
	PrecStd	8.0	2.5	2.4	1.9	1.7	1.6	1.1	0.8	1.7	3.3	4.2	3.2	2.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.9	44.7	51.8	67.0	73.3	80.1	78.4	78.5	67.9	55.1	49.0	47.4
		MCWB	43.2	40.3	43.9	50.9	54.5	60.7	61.1	60.8	54.9	49.7	45.8	44.6
	2%	DB	42.9	42.4	46.2	57.8	68.6	73.3	72.0	71.7	62.5	52.2	46.1	43.8
		MCWB	41.1	39.6	40.5	45.5	53.0	58.4	60.2	58.9	53.1	48.3	43.4	41.9
	5%	DB	40.7	40.5	43.3	53.1	64.6	68.0	68.4	67.3	58.8	50.5	44.1	41.5
		MCWB	39.1	38.0	38.8	43.2	51.9	56.7	59.3	58.2	52.5	47.3	41.7	39.7
	10%	DB	39.0	38.5	41.1	49.6	60.6	64.3	65.1	63.5	56.3	49.2	41.8	39.0
		MCWB	37.6	36.4	37.4	42.1	50.3	55.3	57.9	57.0	52.1	46.3	39.8	37.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.9	41.8	45.4	51.4	56.8	61.7	63.1	62.3	57.4	51.4	46.3	44.3
		MCDB	45.4	43.5	50.0	64.9	69.2	76.6	72.8	75.3	62.2	53.2	48.5	46.1
	2%	WB	41.2	40.2	41.9	46.5	54.7	59.5	61.3	60.2	55.3	49.8	44.1	42.2
		MCDB	42.7	42.0	45.6	55.0	65.7	70.7	70.3	69.0	60.0	51.5	46.0	44.0
	5%	WB	39.2	38.4	39.6	44.4	52.9	57.6	59.7	58.9	53.9	48.1	42.0	39.8
		MCDB	40.6	40.1	42.3	50.9	62.3	66.0	67.2	65.5	57.4	50.1	43.8	41.2
	10%	WB	37.6	36.8	38.0	42.9	51.2	55.9	58.3	57.6	52.7	46.5	39.9	37.3
		MCDB	39.0	38.3	40.4	48.4	59.1	62.9	64.3	62.3	55.6	48.6	41.6	38.5
Mean Daily Temperature Range	5% DB	MDBR	8.4	9.9	10.0	12.8	14.3	12.4	10.4	10.6	9.3	8.3	7.5	8.0
		MCDBR	7.8	9.9	12.2	20.4	23.4	22.6	19.5	20.6	15.8	9.8	8.3	8.7
		MCWBR	6.8	7.5	8.0	11.4	11.7	10.6	9.6	9.9	8.8	6.6	6.7	7.4
	5% WB	MCDBR	7.9	8.3	10.2	17.0	19.5	18.6	17.1	16.2	11.4	7.8	7.5	8.5
		MCWBR	6.9	6.9	7.3	10.2	10.5	9.2	8.9	8.5	7.0	6.2	6.5	7.4
Clear-Sky Solar Irradiance	taub	0.288	0.299	0.331	0.365	0.381	0.416	0.403	0.381	0.343	0.315	0.277	0.255	
	taud	1.997	2.104	2.036	1.999	2.070	2.087	2.208	2.291	2.360	2.217	2.022	1.960	
	Ebn at Noon	147	220	248	261	267	258	259	257	249	213	149	114	
	Edh at Noon	17	25	37	46	46	46	40	34	27	23	16	12	
All-Sky Solar Radiation	RadAvg	123	360	775	1291	1706	1638	1464	1132	727	350	160	78	
	RadStd	11	55	85	131	154	107	122	120	82	44	20	7	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	+1.48	+1.04	N/A	N/A	N/A	+9
(47)	Regional (10 neighbors)	N/A	N/A	+3.13	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page